

Quantum Field Theory — Lecture 2

IMAPP lecture notes

Spring 2026

1 Pictures of time evolution

With a time-independent Hamiltonian, the evolution operator is

$$U(t, t_0) = \exp\left[-\frac{i}{\hbar}H(t - t_0)\right].$$

In the Schrödinger picture states evolve and operators are fixed. In the Heisenberg picture,

$$|\psi\rangle_H = U^\dagger |\psi(t)\rangle_S, \quad A_H(t) = U^\dagger A_S U,$$

so the state is time independent and operators carry the time dependence.

For perturbation theory, split the Hamiltonian into a solvable part and an interaction,

$$H = H_0 + H_{\text{int}}.$$

The interaction picture evolves states with H_{int} and operators with H_0 :

$$|\psi(t)\rangle_I = U_0^\dagger(t, t_0) |\psi(t)\rangle_S, \quad A_I(t) = U_0^\dagger A_S U_0.$$

The operator equation is

$$i\hbar \frac{dA_I}{dt} = [A_I, H_0] + i\hbar \left(\frac{\partial A}{\partial t}\right)_I.$$

2 The quantum harmonic oscillator

In dimensionless variables, $[X, P] = i$, and the ladder operators are

$$a = \frac{1}{\sqrt{2}}(X + iP), \quad a^\dagger = \frac{1}{\sqrt{2}}(X - iP), \quad [a, a^\dagger] = 1.$$

The number operator $N = a^\dagger a$ satisfies

$$[N, a] = -a, \quad [N, a^\dagger] = a^\dagger.$$

Its eigenstates obey

$$N|n\rangle = n|n\rangle, \quad a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle.$$

The state $|0\rangle$ is the vacuum, defined by $a|0\rangle = 0$. The ladder operators remove or add one quantum of excitation.

3 From a discrete string to a field

A discrete chain of masses and springs has a finite set of coordinates $q_i(t)$. Taking the continuum limit sends the lattice spacing $a \rightarrow 0$ while keeping the mass density and elastic modulus fixed. The discrete variables become a displacement field

$$q_i(t) \longrightarrow \phi(x, t).$$

The action becomes an integral over spacetime,

$$S[\phi] = \int dt \int dx \mathcal{L}(\phi, \partial_t \phi, \partial_x \phi).$$

For a simple string, the equation of motion takes the wave-equation form

$$\partial_t^2 \phi - v^2 \partial_x^2 \phi = 0.$$

The important conceptual step is that a field theory has a continuous set of dynamical variables, one for each point in space.

4 Detailed derivation of the oscillator

The action of the number operator on an eigenstate gives

$$[N, a]|n\rangle = N(a|n\rangle) - a(N|n\rangle) = N(a|n\rangle) - n(a|n\rangle) = -a|n\rangle.$$

Therefore

$$N(a|n\rangle) = (n-1)(a|n\rangle).$$

The state $a|n\rangle$ is an eigenstate of N with eigenvalue $n-1$:

$$a|n\rangle = \alpha|n-1\rangle.$$

The normalization is determined by

$$|\alpha|^2 = \langle n-1|\alpha^* \alpha|n-1\rangle = \langle n|a^\dagger a|n\rangle = \langle n|N|n\rangle = n.$$

Choosing $\alpha = \sqrt{n}$ gives

$$a|n\rangle = \sqrt{n}|n-1\rangle.$$

For the vacuum state,

$$a|0\rangle = 0.$$

The operator a is therefore the annihilation operator: it removes one unit of N , i.e. one quantum of energy $\hbar\omega$. Conversely,

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

is the creation operator.

5 From discrete variables to a field

A field theory is a theory with a continuum of variables, rather than a set of discrete variables. Consider an elastic string modelled as an infinite chain of point masses attached by identical springs. The set

$$\{\varphi_i \mid i \in \{1, \dots, N\}\}$$

describes a possible state of the discrete system.

For short-range interactions, the Hamiltonian is

$$H = \sum_i \left[\frac{1}{2} m \dot{\varphi}_i^2 + \frac{1}{2} k (\varphi_{i+1} - \varphi_i)^2 \right].$$

Factoring the lattice spacing a gives

$$H = a \sum_i \left[\frac{1}{2} \frac{m}{a} \dot{\varphi}_i^2 + \frac{1}{2} k a \left(\frac{\varphi_{i+1} - \varphi_i}{a} \right)^2 \right].$$

The corresponding Lagrangian is

$$L = a \sum_i \left[\frac{1}{2} \frac{m}{a} \dot{\varphi}_i^2 - \frac{1}{2} k a \left(\frac{\varphi_{i+1} - \varphi_i}{a} \right)^2 \right].$$

The Euler–Lagrange equation is

$$\frac{\partial L}{\partial \varphi_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\varphi}_i} = 0,$$

which gives

$$\frac{m}{a} \ddot{\varphi}_i - k a \frac{\varphi_{i+1} - 2\varphi_i + \varphi_{i-1}}{a^2} = 0.$$

5.1 Continuum limit

The continuum limit is taken as $a \rightarrow 0$ while

$$\frac{m}{a} \longrightarrow \mu, \quad k a \longrightarrow \chi, \quad i a \longrightarrow x.$$

Here μ is the mass density, χ is Young’s modulus, and x is the position along the string. The discrete variable becomes the displacement field

$$\varphi_i(t) \longrightarrow \varphi(x, t).$$

A discrete set of variables becomes a continuous set of variables, assigning a physical quantity to every space-time point.

The difference is expanded as

$$\varphi_{i+1}(t) - \varphi_i(t) = \varphi(x + a, t) - \varphi(x, t) = \frac{\partial \varphi}{\partial x} a + O(a^2).$$

The sum becomes an integral:

$$a \sum_i \longrightarrow \int_0^L dx.$$

Thus

$$S = \int_{t_1}^{t_2} dt \int_0^L dx \mathcal{L},$$

with Lagrangian density

$$\mathcal{L} = \frac{1}{2} \mu (\partial_t \varphi)^2 - \frac{1}{2} \chi (\partial_x \varphi)^2.$$

The notes write the continuum equation of motion as

$$-\mu \left(\frac{\partial \varphi}{\partial t} \right)^2 + \chi \left(\frac{\partial \varphi}{\partial x} \right)^2 = 0.$$

In general,

$$\mathcal{L} = \mathcal{L}(\partial_t\varphi, \partial_x\varphi; t).$$

The action is local because it is an integral over time and space of a Lagrangian density depending on the value of the field and its derivatives at one point. The Lagrangian density is a polynomial. The Lagrangian formalism treats time and space on the same footing, as required by relativity. The Hamiltonian makes time explicit while space is disentangled.

Summary

The interaction picture separates free evolution from interactions, while the harmonic oscillator shows how quantized excitations arise. The continuum limit replaces discrete coordinates by fields and sums by spatial integrals.